

SymPy Bug Card

Bug 22: solveset returns 1 for $\arccos(x) = 2\pi$

Task. Solve the principal inverse-cosine equation over the complex domain.

$$\text{solveset}(\text{Eq}(\arccos(x), 2\pi), x, \mathbb{C})$$

SymPy's answer.

$$\text{solveset}(\text{Eq}(\arccos(x), 2\pi), x, \mathbb{C}) = \{1\}.$$

Correct answer.

$$\emptyset$$

Explanation. Substituting SymPy's returned value gives $\arccos(1) - 2\pi = -2\pi$, so 1 is not a solution. The error comes from treating $\cos$ as a globally valid inverse of the principal branch $\arccos$, even though $\arccos(\cos(2\pi)) = 0$, not 2π .

Likely root cause. The diagnosis points to `sympy/solvers/solveset.py:_invert_complex`, where the generic inverse-function branch applies `$\arccos(x).inverse() == \cos$` without a principal-range condition and later checks only definedness. This is a new instance of a known failure mode.

Artifact (GitHub). [bug-report/bug-022-solveset-returns-1-for-arccos-x-2-pi/](#)